Metadata-Assisted Retrieval and Rule-Based Extraction for Structured IPL Question Answering

A Personal Research Report

Author: Mridul Sharma
Project date: October 2026
Scope: Experimental study for personal learning and documentation

This report documents a completed personal experiment. It is not presented as a journal submission or as evidence of general-purpose hallucination reduction.

Abstract

On a frozen set of 180 template-generated questions about historical Indian Premier League (IPL) records, deterministic rule-based extraction achieved 99.44% exact answer accuracy with fixed BM25 evidence and 100.00% with metadata-assisted evidence, compared with 55.56% and 56.11% for the corresponding FLAN-T5-small RAG pipelines. The study compares five pipelines: a no-retrieval generator baseline, fixed BM25 RAG, metadata-assisted RAG, fixed BM25 with rule-based extraction, and metadata-assisted retrieval with the same extractor. The sample contains 60 batter-run, 60 innings-total, and 60 match-winner questions. Metadata assistance increased generative accuracy by one answer relative to fixed BM25 RAG ($p = 1.00$); fixed BM25 plus extraction outperformed fixed BM25 plus generation on 79 discordant questions (exact $p \approx 3.31e\text{-}24$). Target passages were retrieved for 179/180 questions with fixed BM25 and 180/180 with metadata-assisted selection. These results show that, for predictable questions over consistently formatted records, schema-specific extraction can be more reliable than free-form generation. The benchmark is narrow and template-generated, the extractor is tailored to its passage format, and exact answer accuracy does not directly measure evidence support or hallucination.

Keywords: retrieval-augmented generation; BM25; rule-based extraction; structured question answering; evidence grounding; IPL; language models

1. Introduction

Retrieval-augmented generation (RAG) combines external evidence with a language model's parametric knowledge. Yet a pipeline can retrieve the relevant passage and still return an incorrect answer if the generator selects the wrong field, copies a nearby value, or produces an incomplete response. This distinction is especially visible in structured question answering, where a question may request a single value that is explicitly represented in a record.

This personal experiment examines that distinction using historical IPL records transformed into structured passages and template-generated short-answer questions. It compares retrieval policies and answer-production methods rather than proposing a new trained architecture. The metadata-assisted policy uses hand-written, task-specific rules; the extraction method uses regular expressions and deterministic rules tailored to the passage format.

The contributions are limited to this benchmark and its saved evaluation artifacts:

- A five-pipeline comparison on a frozen sample of 180 questions, with 60 questions in each of three categories.
- A separate account of target-passage hit rate and exact answer accuracy, including diagnostics for match-winner generation errors.
- A scoped comparison of free-form generation with schema-specific extraction, with statistical uncertainty and validity limitations reported explicitly.

This report does not claim general-purpose hallucination reduction. It treats answer accuracy, retrieval success, and evidence support as distinct outcomes.

2. Literature Review

2.1 Retrieval-augmented generation

Lewis et al. (2020) describe retrieval-augmented generation as combining a language model's parametric knowledge with external, non-parametric memory. In a practical pipeline, retrieval and answer production are separate stages: a retriever must find useful evidence, and the generator must use that evidence correctly. This project measures target-passage retrieval separately from exact answer correctness.

2.2 Lexical retrieval and fixed retrieval policies

BM25 is a lexical ranking function that scores candidate documents using query-term matching with adjustments for term frequency, document length, and term rarity (Robertson & Zaragoza, 2009). Its transparency and low computational cost make it a practical baseline for this project. A fixed top-k policy applies the same retrieval rule to each question, while a task-specific policy can inspect a bounded candidate pool and apply known metadata checks.

2.3 Adaptive, reflective, and corrective RAG

Adaptive-RAG (Jeong et al., 2024) uses a learned classifier to select retrieval strategies according to question complexity. Self-RAG (Asai et al., 2024) trains a model to retrieve when needed and reflect on evidence relevance and answer support. Corrective RAG (Yan et al., 2024) assesses retrieval quality and takes corrective actions when retrieved material may be inadequate. The method in this report is simpler: deterministic, task-specific filtering of BM25

candidates using known fields such as date, batter, innings, and batting team. It does not learn a retrieval policy and does not implement self-reflection or corrective generation.

2.4 Structured extraction versus generation

When records follow a stable schema, a deterministic extractor can identify an explicitly represented field without asking a language model to restate it. This is not equivalent to general natural-language reasoning. Extraction may be reliable when patterns are predictable, but it can be brittle when schemas change, evidence is incomplete, or a question requires synthesis across records. The present comparison is therefore between free-form generation and schema-specific extraction on structured data.

2.5 Scope of the project

The practical question is whether errors in this pipeline arise because the target passage is not retrieved or because the answer-production step fails despite relevant evidence being available. The experiment addresses this question through paired evaluation of five pipelines and inspection of selected errors. Its value is diagnostic and educational, not a claim of a new adaptive RAG architecture.

3. Research Questions

1. RQ1: On the frozen 180-question sample, how does exact answer accuracy for FLAN-T5-small change when fixed BM25 retrieval is added compared with no retrieved evidence?
2. RQ2: Does the implemented metadata-assisted retrieval policy improve exact answer accuracy over fixed BM25 retrieval for the same generator?
3. RQ3: On the same questions, how does deterministic rule-based extraction compare with FLAN-T5-small generation when paired with fixed BM25 evidence?
4. RQ4: What do target-passage hit rates and the recorded error diagnostics support, and not support, about interpreting exact answer accuracy as evidence of hallucination reduction?

4. Data and Experimental Design

4.1 Source and benchmark construction

The benchmark was constructed from historical IPL match data distributed by Cricsheet. Structured match records were transformed into passages and template-generated short-answer questions. The question types are batter_runs, innings_runs, and match_winner. The frozen evaluation contains 180 unique questions, evenly split into 60 per category. Questions were sampled from an existing held-out test partition using the recorded seed 20261028. The sample excludes the earlier 60-question development pilot, but it is not an independent external benchmark.

The evaluation artifact records question IDs, gold answers, target passage IDs, target-passage existence, expected split membership, and target evidence text. All 180 questions were marked as belonging to the test split, and their target passages were present in the evaluation data. These checks support internal consistency of the frozen sample, not external validity.

4.2 Methods compared

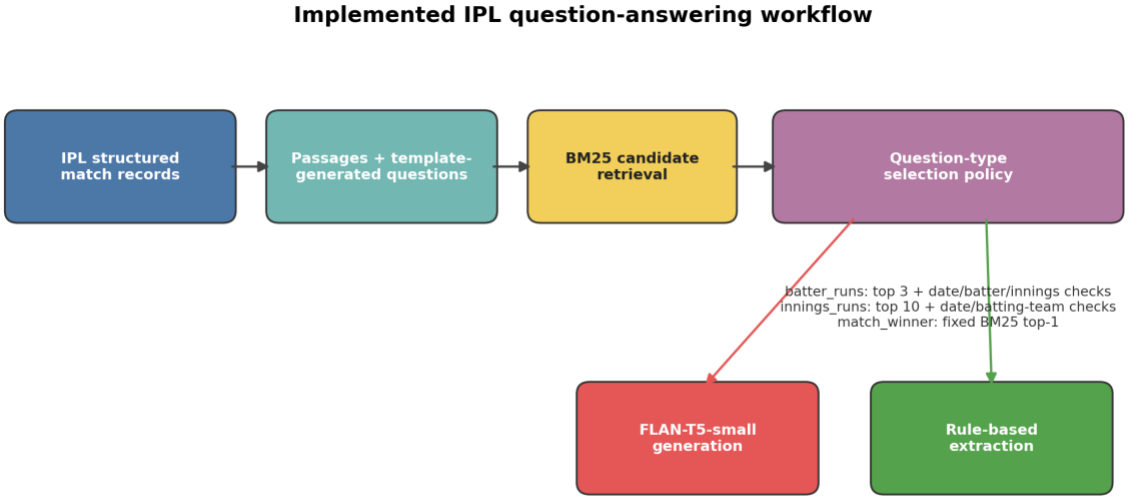


Figure 1. Implemented workflow and task-specific evidence-selection policy.

Method	Implementation
M1. No retrieval + FLAN-T5-small	The generator answers without retrieved benchmark evidence.
M2. Fixed BM25 RAG + FLAN-T5-small	BM25 retrieves the top-ranked passage, which is supplied to the generator.
M3. Metadata-assisted RAG + FLAN-T5-small	A task-specific metadata policy selects evidence before the same generator answers.
M4. Fixed BM25 + deterministic extraction	A rule-based extractor produces an answer from fixed-BM25 evidence.
M5. Metadata-assisted retrieval + deterministic extraction	The same extractor is paired with metadata-assisted evidence selection.

The generator comparison holds the named generator constant between the two RAG methods, but the comparison between generation and extraction is not capacity-matched: FLAN-T5-small is a general-purpose text generator, while the extractor is hand-written for this benchmark's passage templates. This is a fair comparison of the implemented pipelines, not a controlled comparison of model intelligence or general-purpose reasoning.

4.3 Metadata-assisted retrieval policy

Question type	Candidate pool and rule
batter_runs	Inspect the top 3 BM25 candidates; apply date, batter, and innings checks.
innings_runs	Inspect the top 10 BM25 candidates; apply date and batting-team checks.
match_winner	Use fixed BM25 top-1; no additional metadata-based selection step.

The candidate-pool sizes and rules differ by question type. The policy is tailored to the benchmark's known categories and should not be described as a general learned adaptive retrieval strategy.

4.4 Generation and deterministic extraction

The generation model was google/flan-t5-small. Decoding was deterministic, using do_sample=False, num_beams=1, and max_new_tokens=32. The extraction pipeline used the regex/rule-based function extract_answer_from_evidence_v2. It relies on the stable structure of retrieved passages and is not a general-purpose natural-language reasoning method.

Component	Implemented procedure
Evidence selection for batter_runs	Retrieve top 3 BM25 candidates and check date, batter, and innings metadata.
Evidence selection for innings_runs	Retrieve top 10 BM25 candidates and check date and batting-team metadata.
Evidence selection for match_winner	Use fixed BM25 top-1 without additional metadata filtering.

Generation	Pass selected evidence to google/flan-t5-small with do_sample=False, num_beams=1, max_new_tokens=32.
Extraction	Apply extract_answer_from_evidence_v2, a regex/rule-based extractor tailored to the structured passage format.

Pseudocode summary: (1) identify the question category; (2) retrieve BM25 candidates; (3) for batter_runs, inspect top 3 and check date/batter/innings; for innings_runs, inspect top 10 and check date/batting team; for match_winner, retain fixed top-1; (4) either pass selected evidence to the fixed FLAN-T5-small configuration or apply extract_answer_from_evidence_v2; (5) compare the output with the gold answer using exact match.

4.5 Metrics and statistical analysis

The primary metric was exact benchmark answer correctness, calculated as correct answers divided by 180 and reported as a percentage. Wilson 95% confidence intervals were calculated for overall accuracy. Because all methods were evaluated on the same frozen questions, selected pairwise comparisons used exact McNemar tests on paired correctness outcomes. Retrieval quality was measured separately through target-passage hit rates. Error diagnostics characterized incorrect match-winner generation outputs.

Exact match is suitable for short numeric and categorical answers but does not measure semantic equivalence, completeness, calibration, or whether an answer is entailed by evidence. Accordingly, this report uses “answer accuracy” rather than treating the accuracy scores as direct hallucination rates.

5. Results

5.1 Overall answer accuracy

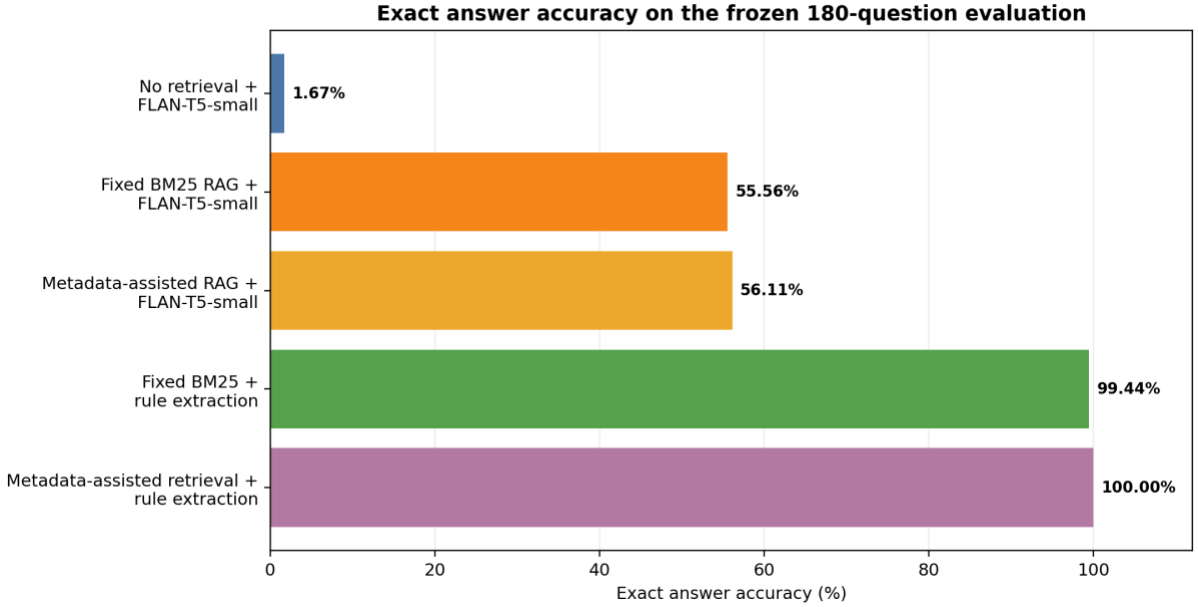


Figure 2. Exact answer accuracy for all five methods on the frozen evaluation.

The no-retrieval baseline answered 3/180 questions correctly (1.67%). Fixed BM25 RAG answered 100/180 (55.56%), while metadata-assisted RAG answered 101/180 (56.11%). Deterministic extraction reached 179/180 (99.44%) with fixed BM25 evidence and 180/180 (100.00%) with metadata-assisted evidence.

Method	Correct / N	Accuracy	Wilson 95% CI
No retrieval + FLAN-T5-small	3/180	1.67%	0.57–4.78%
Fixed BM25 RAG + FLAN-T5-small	100/180	55.56%	48.26–62.62%
Metadata-assisted RAG + FLAN-T5-small	101/180	56.11%	48.81–63.16%
Fixed BM25 + rule extraction	179/180	99.44%	96.92–99.90%
Metadata-assisted retrieval + rule extraction	180/180	100.00%	97.91–100.00%

Interpretation: adding fixed BM25 retrieval coincided with a large increase in generator accuracy relative to the no-retrieval baseline. The metadata policy changed generative accuracy by one answer, while the schema-specific extractor achieved near-ceiling accuracy on this benchmark.

5.2 Accuracy by question category

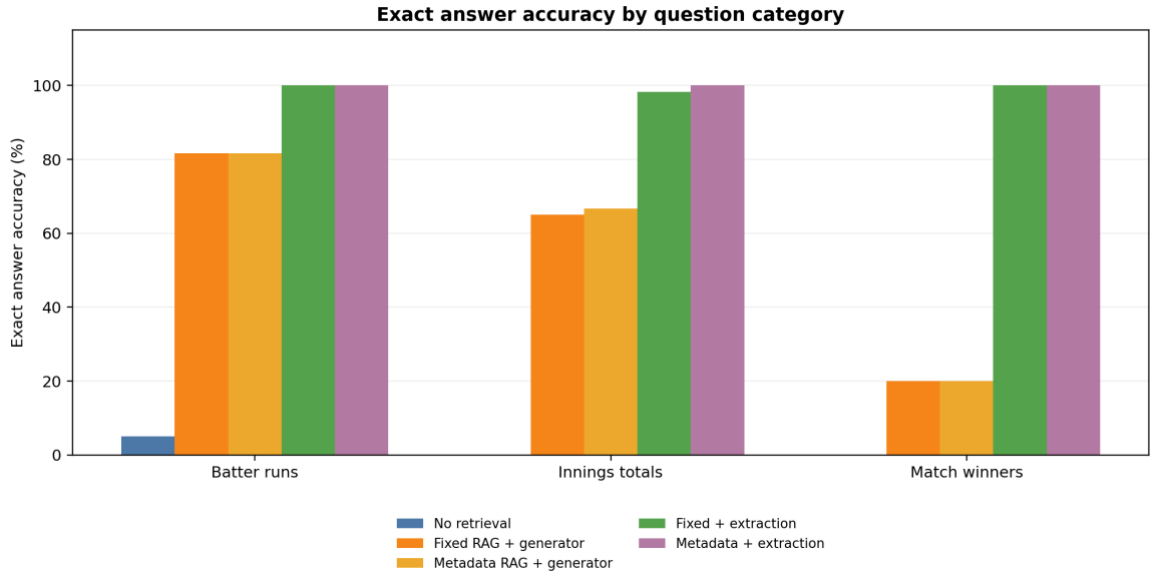


Figure 3. Exact answer accuracy by category; each category contains 60 questions.

Question type	No retrieval	Fixed RAG + generator	Metadata RAG + generator	Fixed + extraction	Metadata + extraction
Batter runs	3/60 (5.00%)	49/60 (81.67%)	49/60 (81.67%)	60/60 (100.00%)	60/60 (100.00%)
Innings totals	0/60 (0.00%)	39/60 (65.00%)	40/60 (66.67%)	59/60 (98.33%)	60/60 (100.00%)
Match winners	0/60 (0.00%)	12/60 (20.00%)	12/60 (20.00%)	60/60 (100.00%)	60/60 (100.00%)

Interpretation: generation accuracy varied by question category, with both RAG generators performing best on batter-run questions and weakest on match-winner questions. Both extraction methods answered all batter-run and match-winner questions correctly; the fixed-evidence extractor missed one innings-total question.

5.3 Confidence intervals and paired tests

The Wilson intervals quantify uncertainty around the observed accuracy estimates. The fixed BM25 generator's interval was 48.26%–62.62%, while the metadata-assisted generator's interval was 48.81%–63.16%. These intervals are close, consistent with the small observed difference.

Method A	Method B	A-only correct	B-only correct	Discordant pairs	Exact p
Fixed BM25 RAG + FLAN-T5-small	Metadata-assisted RAG + FLAN-T5-small	0	1	1	1
Fixed BM25 + rule extraction	Metadata-assisted retrieval + rule extraction	0	1	1	1
Fixed BM25 RAG + FLAN-T5-small	Fixed BM25 + rule extraction	0	79	79	3.31e-24

Interpretation: the first two paired comparisons each had one discordant question and p = 1.00 in the report's prose; these results do not establish equivalence. Fixed BM25 plus extraction outperformed fixed BM25 plus generation on 79 discordant questions, with none favouring generation (exact p ≈ 3.31e-24).

5.4 Retrieval diagnostics

Question type	Fixed BM25 target hits	Metadata-assisted target hits
Batter runs	60/60 (100.00%)	60/60 (100.00%)
Innings totals	59/60 (98.33%)	60/60 (100.00%)
Match winners	60/60 (100.00%)	60/60 (100.00%)
Overall	179/180 (99.44%)	180/180 (100.00%)

Interpretation: the only category-level retrieval difference was for innings totals, where metadata-assisted selection recovered the target passage missed by fixed BM25. Both methods retrieved the target passage for every batter-run and match-winner question.

5.5 Retrieval success versus answer correctness

Target-passage retrieval and answer correctness are distinct outcomes. Fixed BM25 retrieved the target passage for 179/180 questions (99.44%), while metadata-assisted selection did so for 180/180 (100.00%). Nevertheless, the RAG generators answered only 55.56% and 56.11% correctly overall. For the 48 incorrect match-winner outputs examined, the target winner was present in the evidence. These diagnostics support the narrow conclusion that many observed generation errors occurred despite relevant evidence being available; they do not establish that every error in every category was conditional on a correct retrieval or that all wrong answers were hallucinations.

5.6 Match-winner generation errors

The match-winner diagnostics record 48 incorrect outputs among 60 questions for each RAG generation method. In the 48 cases examined, the target winner was present in the evidence.

Recorded error category	Count	Share of 48 errors
Match ID copied as answer	14	29.17%
Zero output	12	25.00%
Team name, but incorrect winner	10	20.83%
Player of the match copied as answer	7	14.58%
Other incorrect response	5	10.42%

Interpretation: the most frequent recorded category was copying the match ID as the answer, followed by zero output and an incorrect team name. These descriptive categories characterize errors in this benchmark; they are not a comprehensive taxonomy of hallucinations.

6. Discussion

6.1 Answers to the research questions

RQ1: Fixed BM25 retrieval increased FLAN-T5-small exact answer accuracy from 1.67% without retrieval to 55.56% with fixed BM25 RAG on this sample. This is an observed difference on the benchmark, not a claim that retrieval guarantees factual answers.

RQ2: Metadata-assisted RAG reached 56.11%, compared with 55.56% for fixed BM25 RAG. The paired comparison had one discordant question and $p = 1.00$. The sample therefore provides little evidence that the implemented metadata policy changed generation accuracy; it does not prove equivalence.

RQ3: Fixed BM25 plus deterministic extraction reached 99.44%, compared with 55.56% for fixed BM25 plus generation. There were 79 discordant questions favouring extraction and none favouring generation (exact $p \approx 3.31e\text{-}24$). This comparison concerns a benchmark-specific rule-based extractor versus FLAN-T5-small, not a capacity-matched comparison of general-purpose systems.

RQ4: The target-passage hit rates were 179/180 for fixed BM25 and 180/180 for metadata-assisted selection. The match-winner diagnostics found the target winner in the evidence for the 48 incorrect outputs examined. These findings show that retrieval success does not guarantee answer correctness, but the study does not directly measure hallucination or evidence entailment.

6.2 Why metadata assistance yielded a null result

The observed difference between the two RAG generators was one answer, and the paired test was not significant ($p = 1.00$). The policy's design helps explain why a larger effect was not observed: match-winner questions use fixed BM25 top-1 without an additional metadata-based selection step, even though match winners were the weakest category for generation. The policy also changed target-passage retrieval by only one question overall. These are explanations grounded in the implemented policy and observed counts, not proof that metadata checks cannot help on other tasks.

6.3 Why ceiling extraction accuracy is expected here

The extraction result is consistent with a benchmark in which questions are template-generated from structured records and passages follow a stable format. The extractor is explicitly tailored to those passage patterns and to the expected

answer fields. Under these conditions, a rule-based function can map a known pattern to a field without generating surrounding text. The 100.00% observed score is therefore a result for this benchmark and implementation, not evidence that extraction is universally reliable or that the method would retain its performance on paraphrased, ambiguous, incomplete, or unstructured inputs.

6.4 Hypotheses about generator errors

The following are hypotheses, not findings established by the experiment:

- Prompt/task-format hypothesis: the generator may not have been sufficiently constrained to return only the requested field, which could contribute to copying a match ID or player-of-the-match value.
- Model-capacity hypothesis: FLAN-T5-small may have difficulty consistently selecting the correct field from passages containing several nearby values. The experiment did not compare model sizes, so this remains untested.
- Output-format hypothesis: the zero-output category may reflect a mismatch between the model's learned response patterns and the short-answer format. The saved error categories alone do not identify the cause.

Testing these hypotheses would require controlled prompt variants, larger models, and direct inspection or annotation of outputs; none of those experiments is claimed here.

6.5 What the results say about hallucination

This experiment studies factual answer errors in a retrieval-grounded workflow, but it does not operationalize hallucination as a separate measured outcome. Exact answer correctness, retrieval success, and evidence support are different: an answer may be correct without being grounded, and an incorrect answer may arise from retrieval, field selection, extraction, formatting, or generation. The defensible conclusion is that schema-specific extraction improved exact-match accuracy on this structured benchmark. A direct claim about hallucination reduction would require explicit evidence-support annotation or a validated grounding metric on more varied data.

6.6 Fairness and interpretation of the comparison

The comparison answers a practical engineering question: which of these implemented pipelines produces the expected answer on this benchmark? It is not a balanced contest between methods with comparable development effort. The generator is a small general-purpose model, while the extractor contains hand-written rules tailored to the known schema. That asymmetry is important to interpretation and is a reason to treat the extraction result as a strong benchmark-specific baseline rather than a general claim about language models.

7. Threats to Validity

Threat	Why it matters	How it affects interpretation	What would address it
Domain scope	All questions concern historical IPL cricket records.	Results may not transfer to other domains or data formats.	Evaluate on additional domains and datasets.
Template dependence	Questions are generated from structured records and cover three short-answer categories.	Predictable wording and schema patterns may favour rule-based extraction.	Add paraphrased questions, varied formats, and less templatic inputs.
Sample size and independence	The frozen sample contains 180 questions from an existing held-out partition, not an independent external benchmark.	Confidence intervals describe this sample's uncertainty but do not establish external validity.	Use larger, independently constructed evaluation sets.
Method specificity	Metadata-assisted retrieval uses hand-written rules and category-specific candidate-pool sizes.	The method is not a learned adaptive retrieval policy; effects may depend on these rules.	Compare with other retrieval policies under a pre-specified protocol.
Extractor specificity	The extractor relies on known passage structure and regex/rule patterns.	High accuracy may depend on stable templates and may fall when formatting changes.	Test schema variation, malformed records, and out-of-pattern questions.
Metric limitations	Exact match does not measure entailment, evidence attribution, completeness, or calibration.	Accuracy cannot be treated as a direct hallucination rate.	Annotate evidence support and evaluate grounding separately.
Model scope	The generator comparison uses only FLAN-T5-small with one deterministic decoding configuration.	The results do not show how other model sizes or prompts would perform.	Compare model sizes and controlled prompt variants.
Statistical scope	Paired tests were run for selected comparisons; a non-significant result is not proof of equivalence.	The metadata comparison's null result should not be interpreted as equivalence.	Pre-specify additional comparisons and evaluate more questions or repeated samples.

8. Future Work

The following work is prioritized by how directly it addresses the present benchmark's limitations. These are proposals, not completed experiments.

5. Paraphrased questions: evaluate whether the methods remain reliable when questions are not generated from the same templates.
6. Unanswerable questions: include questions for which the evidence does not contain the requested answer, then measure abstention and unsupported-answer behaviour.
7. Larger models and better prompts: compare additional model sizes and controlled prompts to test the hypotheses about field selection and output format.
8. Hybrid extractor-plus-generator fallback: use deterministic extraction when a recognized schema pattern is present and a generator fallback when it is not, with explicit evaluation of fallback decisions.
9. Dense retrieval: compare lexical BM25 with dense retrieval under the same frozen or independently held-out evaluation protocol.
10. Direct evidence-support annotation: annotate whether each answer is supported by its retrieved evidence so grounding can be evaluated separately from exact match.
11. Multiple seeds: repeat sample construction or relevant stochastic components across multiple seeds where applicable, and report variability without mixing development and final evaluation results.

9. Conclusion

On a frozen sample of 180 template-generated questions about historical IPL records, fixed BM25 RAG increased FLAN-T5-small exact answer accuracy from 1.67% to 55.56%, while metadata-assisted RAG reached 56.11%. The metadata policy changed generator accuracy by one answer and did not yield a significant paired difference ($p = 1.00$). Rule-based extraction achieved 99.44% with fixed BM25 evidence and 100.00% with metadata-assisted evidence; fixed BM25 plus extraction outperformed fixed BM25 plus generation on 79 discordant questions (exact $p \approx 3.31e\text{-}24$). Target-passage hits were 179/180 and 180/180, respectively, showing that successful retrieval did not guarantee correct generation. The scoped takeaway is that a schema-specific extractor can outperform a small generator on predictable structured questions. These results do not establish general hallucination reduction or general superiority over language models.

10. Data, Code, and Reproducibility Notes

The source data is historical IPL match data from Cricsheet (https://cricsheet.org/downloads/). The project repository contains the cleaned notebook and evaluation artifacts, including the frozen question sample, result tables, confidence intervals, paired tests, and diagnostics. The notebook was reported by the project owner to run successfully from a fresh Colab runtime, with tables displayed and no errors. This report relies on the saved artifacts and that reported run; it does not claim an independent execution of the entire notebook during this revision. Data reuse should follow Cricsheet's current terms and attribution requirements.

The earlier 60-question development pilot is separate from the final 180-question evaluation. Its consolidated method summary is a development artifact and is not used in the main results.

References

Asai, A., Wu, Z., Wang, Y., Sil, A., & Hajishirzi, H. (2024). Self-RAG: Learning to retrieve, generate, and critique through self-reflection. International Conference on Learning Representations (ICLR). https://openreview.net/forum?id=hSyW5go0v8

Cricsheet. (n.d.). Downloads: Ball-by-ball cricket data. https://cricsheet.org/downloads/

Jeong, S., Baek, J., Cho, S., Hwang, S. J., & Park, J. C. (2024). Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity. Proceedings of NAACL-HLT 2024, 7036–7050. https://aclanthology.org/2024.naacl-long.389/

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-t., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. Advances in Neural Information Processing Systems, 33. https://arxiv.org/abs/2005.11401

Robertson, S., & Zaragoza, H. (2009). The probabilistic relevance framework: BM25 and beyond. Foundations and Trends in Information Retrieval, 3(4), 333–389. https://doi.org/10.1561/1500000019

Yan, S.-Q., Gu, J.-C., Zhu, Y., & Ling, Z.-H. (2024). Corrective retrieval augmented generation. arXiv:2401.15884. https://arxiv.org/abs/2401.15884

Appendix A. Pilot and frozen evaluation are different

The project contains an earlier 60-question pilot with a different sample and experiment stage. Its summary must not be combined with the frozen 180-question results. The main results and conclusions use only the 180-question frozen evaluation artifacts.

Pilot method	Correct / N	Accuracy
No retrieval + FLAN-T5-small	5/60	8.33%
Fixed BM25 RAG + FLAN-T5-small	33/60	55.00%
Metadata-assisted RAG + FLAN-T5-small	34/60	56.67%
Fixed BM25 + rule-based extraction	59/60	98.33%
Metadata-assisted retrieval + rule-based extraction	60/60	100.00%

The pilot figures are included solely for transparency about the development history. They are not used to estimate final performance.

Appendix B. Artifact audit summary

The available evaluation files were checked for internal consistency before preparing the original report: the frozen question file contains 180 unique IDs; the three question types contain 60 questions each; the confidence-interval table reports five methods over 180 questions; the main summary and question-type table agree with the reported overall scores; and the paired-test artifact contains the three comparisons discussed above. Diagnostic files were reviewed for retrieval hit rates, extraction accuracy, and match-winner error categories. The audit was an inspection of saved artifacts, not a fresh independent re-run of every model prediction.

.